

# Autodesk® Scaleform®

Scaleform 4.3

Scaleform C++ API

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: 2.2

: 2012 6 21

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## 소개 에서의 폰트

Scaleform HTML  
 ,  
 , GFX/SWF , OS FreeType 2  
 ,  
 ,

Scaleform

SWF

Scaleform    Gfx::Translator  
Gfx::FontLib

OS    FreeType2

Scaleform

Scaleform

. Scaleform

Scaleform

UI

ins and

out

TextField

UI

UI

**1.1**

UI

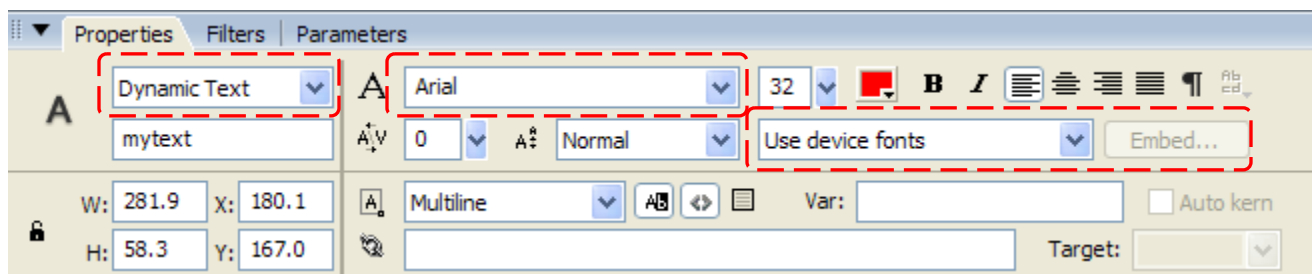
1 -

## 텍스트 필드

Text Tool

Text

Field



•

ActionScript

GFX::Translator

(a)

(b)

Bold Italic

"Arial"

SWF

8

•

•

( )

•

•

" " , OS

SWF

/

, OS

. Scaleform

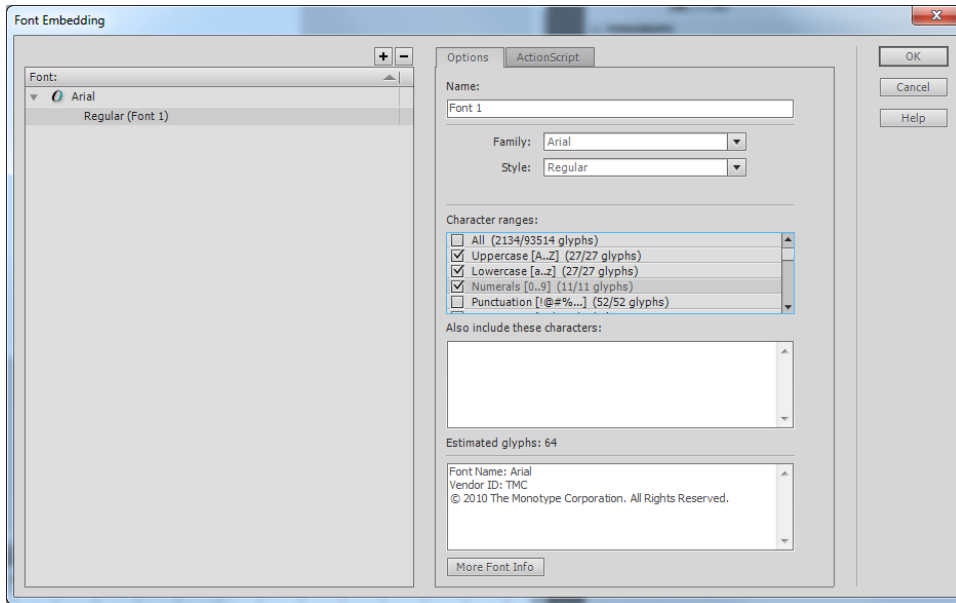
, " " " "

, Embed



## 문자 내장

TextField  
( textfield ).  
(  
Scaleform Gfx::FontLib ). Embed



( " " ). Arial Arial Bold  
footprint

SWF/GFX

. Scaleform

(Xbox 360, PS3, Wii) PC ( , , )

## 내장 폰트를 통한 메모리 사용

Scaleform

| 내장 문자 수  | 비압축 크기   | 런타임 크기  |
|----------|----------|---------|
| 1        | 1 KB     | 450 K   |
| 114 - +  | 12 KB    | 480 K   |
| 596 - +  | 70 KB    | 630 K   |
| 7,583 -  | 2,089 KB | 3,500 K |
| 18,437 - | 5,131 KB | 8,000 K |

"Arial Unicode MS"

500 150K

## 폰트 메모리 사용 제어

1. *UI*

2. *Gfx::FontLib* /

(  
*SWF/GFX*

*Gfx::FontLib*

3.

IME

4. *GfxEport* (-fc )

~ 30%

( 5.3 ).

10%

*Gfx::FontLib*

IME

## 1.2 *UI*

Scaleform

- 1 : –
- 2 : –
- 3 : – Scaleform
- 4 : – Scaleform

UI

UI,

2

- , Gfx::FontLib
- , ( ).
- , SWF/GFX

Scaleform

, 3 : Gfx::FontLib  
Scaleform

FreeType2

Gfx::FontLib

4

Scaleform

. GfxExport

,

.

.

## 제 부 게임 폰트 라이브러리 생성

UI

UI

1.

UI

2.

3.

UI

, Scaleform

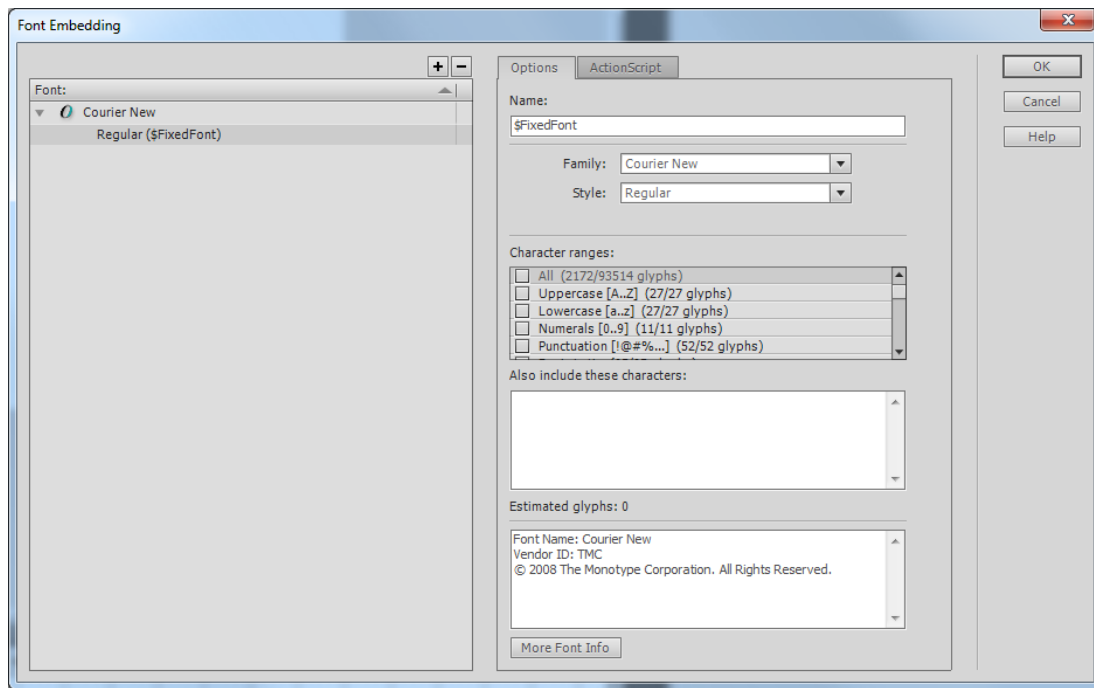
("fonts\_en.swf", "fonts\_kr.swf" ). .

SWF

### 2.1

. TextField

"New Font..."



FLA

,\$FixedFont"

TextField

, "\$ArialGame\*" \*

UI

'\$'

TextField

"Font rendering method"

폰트 기호의 불러오기 및 내보내기

, FLA

/

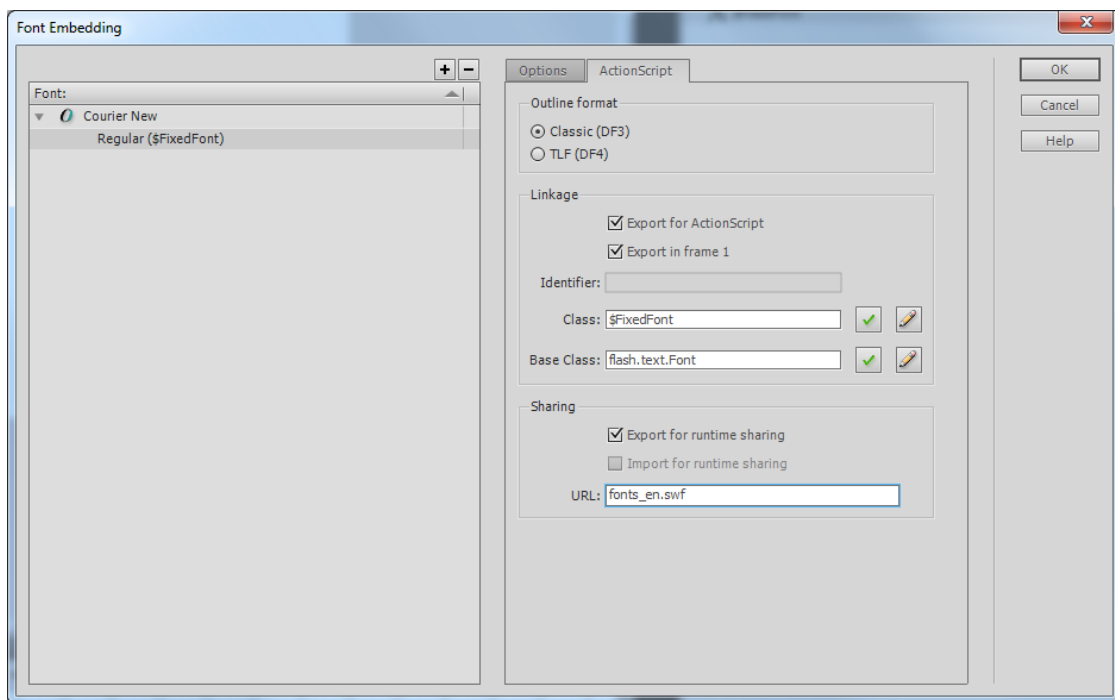
FLA

"Copy"

FLA

"Paste"

"ActionScript"



"OK"

Flash

"ActionScript"

URL

SWF

GFX::FontLib

URL

"fonts\_en.swf"

폰트 대체를 사용하려면

에서 기본 폰트 라이브러리를 설정해야 함을 명심하십시오



```
Loader.SetDefaultFontLibName("fonts_en.swf");
```

```
fonconfig.txt      GfxPlayer  
fontlib "fonts_en.swf"
```

fontlib

, Linkage "Export: \$identifier" Copy/Paste

SWF

. Scaleform

SWF

TextField.htmlText

TextFormat

"Arial"

ActionScript

( ).

## 내보낸 폰트 및 문자 내장

CS4 Flash Studio

₩ " " ₩ " "

243 " " 11,920

, 'gfxfontlib.swf'

Flash CS5

'gfxfontlib.swf'  
( 'fonts\_en.swf', 'fonts\_jp.swf' )

CS4 Flash Studio( )

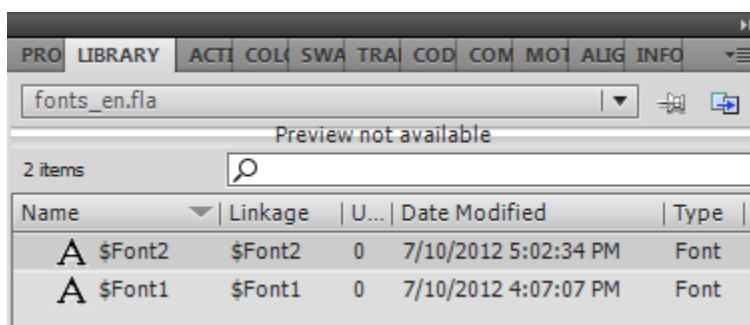
'gfxfontlib.swf'

( , ).

Gfx 4.0

## 2.2 gfxfontlib.swf

,  
 .  
 .  
 , Flash Studio FLA  
 .  
 1. . '\$TitleFont'  
 '\$NormalFont'  
 .  
 2. FLA  
 3. 1  
 4.  
 5.  
 6. fonts\_<lang>.swf ( 'lang' , 'en' – , 'ru'  
 – , 'jp' – , 'cn' – , 'kr' – ).  
 , FontMap  
 ,  
 textfield  
 . FLA  
 ,



Copy/Paste

## 제 부 국제화 접근 선택

UI

1.

2.

3.

default 'fonts\_en.swf'

Gfx::FontLib

Gfx::FontMap

Gfx::Translator

UI

UI

### 3.1

"fonts\_en.swf")

UI

```

    . UI                                     , "fonts_en.swf"
                                     .
                                     Scaleform
    ,
    .

```

```

Loader::SetDefaultFontLibName (const char* filename) C++      'filename'
(
    !)
    .
    .

```

```

Loader.SetDefaultFontLibName("fonts_kr.swf");

```

```

    SWF      CreateInstance
    .
    ,
    .

```

```

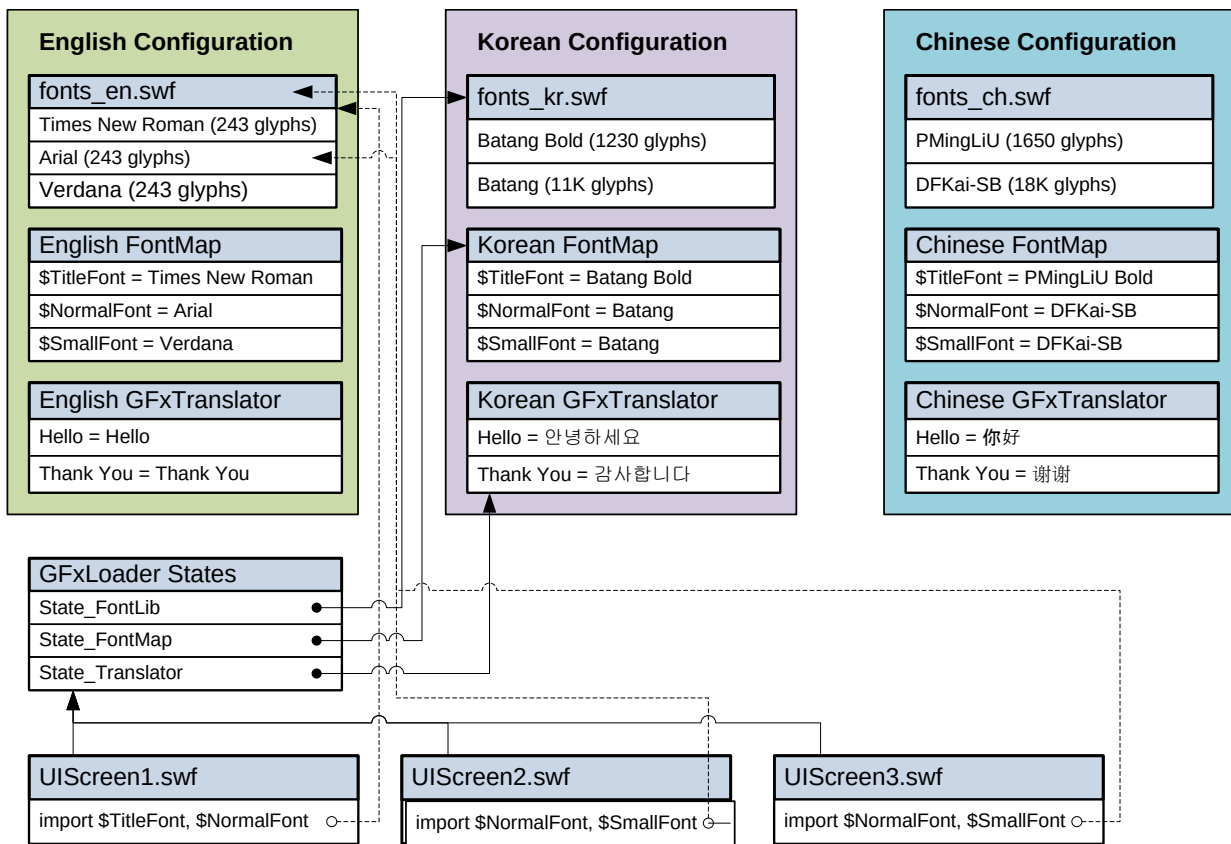
"$NormalFont"

```

```

Scaleform      Gfx::Loader      Gfx::Translator, Gfx::FontLib
Gfx::FontMap      . Gfx::Loader::CreateMovie
    ,
    .
    ,
CreateMovie
    .

```



UIScreen3.swf, UIScreen1.swf, "fonts\_en.swf"

Scaleform Gfx::Loader Gfx::FontLib Gfx::Fontmap

. (1) Loader::SetDefaultFontLibName() , (2)

UI null Gfx::FontLib loader

"fonts\_kr.swf"

UIScreen

## 국제 텍스트 설정

```

        ,
        SWF ( GFX )
        Gfx::FontMap Gfx::Translator
        . Scaleform , "fontconfig.txt"
        .
        .
        1. 1 ( "fonts_en.swf")
        .
        2. Gfx::Translator
        3. . Gfx::FontMap
        .
        4. "fonts_kr.swf"
        .
        , UI
        ,
        1. Gfx::Loader
        Gfx::Translator Translate override
        override ,
        .
        2. Gfx::FontMap Gfx::Loader . MapFont
        . Gfx::FontMap 3
        .
        3. Gfx::FontProvider / Gfx::FontPackParams
        GlyphCacheConfig
        .
        4. Gfx::FontLib Gfx::Loader
        5. (swf ) SetDefaultFontLibName( )
        .
    
```

```

Loader.SetDefaultFontLibName("fonts_en.swf");
Gfx::FontLib , SWF/GFX
Gfx::FontLib::AddFromFile .

```

6. Gfx::Loader::CreateMovie . fontlib

## 에서의 국제화

```

, Scaleform Player "fontconfig.txt"
. SWF/GFX , 'fontconfig.txt'
/fc
, CTRL + N . F2
HUD
8 ASCII UTF-16

```

```

[FontConfig "English"]
fontlib "fonts_en.swf"
map "$TitleFont" = "Times New Roman" Normal
map "$NormalFont" = "Arial " Normal
map "$SmallFont" = "Verdana" Normal

```

```

[FontConfig "Korean"]
fontlib "fonts_kr.swf"
map "$TitleFont" = "Batang" Bold
map "$NormalFont" = "Batang" Normal
map "$SmallFont" = "Batang" Normal
tr "Hello" = "안녕하세요"
tr "Thank You" = "감사합니다"

```

```

[FontConfig "Chinese"]
fontlib "fonts_ch.swf"
map "$TitleFont" = "PMingLiU" Bold
map "$NormalFont" = "DFKai-SB" Normal
map "$SmallFont" = "DFKai-SB" Normal
tr "Hello" = "你好"

```



```
tr "Thank You" = "谢谢"
```

```
, [FontConfig "name"]
```

Scaleform Player HUD

| 설정 설명                       | 의미                                                   |
|-----------------------------|------------------------------------------------------|
| fontlib "fontfile"          | SWF/GFX Gfx::FontLib<br>fontlib                      |
| map "\$UIFont" = "PMingLiU" | UI<br>Gfx::FontMap<br>, \$UIFont "gfxfontlib.swf" UI |
| tr "Hello" = "□好"           |                                                      |

Scaleform

Scaleform

XML

FontConfigParser.h/.cpp

Bin\Data\AS2\Samples\FontConfig and

Bin\Data\AS3\Samples\FontConfig Scaleform 2.x

Scaleform Player "sample.swf" "fontconfig.txt"

## 장치 폰트 에뮬레이션

Gfx::FontMap

. UI , "Arial"

"Verdana"

UI

```

    . Scaleform Player
        , "Arial"
        " "
        " " Gfx::FontLib "font_kr.swf"
        Gfx::FontLib Gfx::FontProviderWin32
        Gfx::FontProviderFT2
    .
    .
    •
    .
    .
    .
    Scaleform UI
    .
    •
    " "
    Scaleform Player ActionScript TextField.antiAliasType
    .
    •
    UI
    Scaleform
    .
    , lib ("fonts_en.swf")
    .
    • ActionScript TextFormat
    . "$TitleFont" "Arial"
    .
    "fonts_en.swf"
    . "fonts_en.swf"
    . UI
    .
    .

```

## 폰트 라이브러리를 생성하기 위한 단계별 가이드

2. fontconfig.txt , SWF .
1. FLA , ActionScript 2.0 3.0 . 'main-app.flaswf' .
2. FLA , ActionScript . 'fonts\_en.swf' .
3. ' , , ' ...' \$Font1 , "Family" "Arial" , "Style" "Regular" .
4. " " . 'Basic Latin' .
5. 'ActionScript' . 'ActionScript' ' , ' 1 ' , ' 'URL' 'fonts\_en.swf' . 'OK' . ActionScript 3.0 Flash Studio ' ...' . 'OK' .
6. 3 ~ 5 , '\$Font2' 'Family' / 'Style' . 'Family' 'Style' , Flash Studio 'Times New Roman' .
7. FLA ActionScript . 'fonts\_kr.flaswf' 'fonts\_ru.flaswf' . 'fonts\_kr.flaswf' .
8. ' , , ' ...' '\$Font1' , "Family" , " " " , "Style" "Regular" .
9. " " . , ' ( )' .

10. 'ActionScript' . 'ActionScript' , ' 1 ,  
' , 'URL' 'fonts\_kr.swf' .  
'OK' . ActionScript 3.0 Flash Studio '  
... 'OK'
11. 8 ~ 10 , '\$Font2' 'Family' ,  
' , .
12. 'fonts\_en.swf' 'fonts\_kr.swf' .
13. main-app.fla . ( 'fonts\_en.swf') , '\$Font1' '\$Font2'  
(Ctrl-C -> ' ).
14. 'main-app.fla' , (Ctrl-V  
-> ' ). ' :'
15. 'main-app.fla' . 'Classic Text', 'Dynamic Text'  
'Family' '\$Font1\*' .  
'\$TEXT1' ( ID ).
16. , '\$Font2\*' , '\$TEXT2' .
17. 'main-app.swf' .
18. swf fontconfig.txt . Unicode UTF-8  
' , , .

```
[FontConfig "English"]
fontlib "fonts_en.swf"
map "$Font1" = "Arial"
map "$Font2" = "Times New Roman"
tr "$TEXT1" = "This is"
tr "$TEXT2" = "ENGLISH!"
```

```
[FontConfig "Korean"]
```

```
fontlib "fonts_kr.swf"
map "$Font1" = "Batang"
map "$Font2" = "BatangChe"
tr "$TEXT1" = "      "
tr "$TEXT2" = "      !"
```

중요 (" " " "), Flash SWF  
 ("Batang" "BatangChe"). fontconfig.txt  
 . SWF *gfxexport* -  
*fntlst* <swfname> \*.lst .

19. ( Unicode UTF-8 !). !  
 GfxPlayer 'main-app.swf' 'This is' 'English!' . Ctrl-  
 N fonts\_kr.swf  
 ( fonts\_kr.swf ).

## 3.2

SWF/GFX  
 "UIScreen.fla" SWF  
 "UIScreen.swf" "UIScreen.swf"  
 .  
 .  
 .  
 .  
 (UI 600K )  
 .  
 UI

UI

( ,

)

UI

IME

.

"

"

,

(CTRL + F11)

.

Scaleform

"Replace string" "

"

.

## 제 부 폰트 소스 설정

HTML

- 1.
2. SWF/GFX
3. Gfx::FontLib SWF/GFX ,
4. Gfx::FontProviderWin32

Gfx::FontLib, Gfx::FontMap

Gfx::FontProvider , Gfx::FontLib Gfx::FontMap

SWF/GFX

Gfx::Loader::SetFontProvider

SWF

SWF

Scaleform

- Gfx::FontProviderWin32 – Win32
- Gfx::FontProviderFT2 – David Turner  
FreeType-2

## 4.1

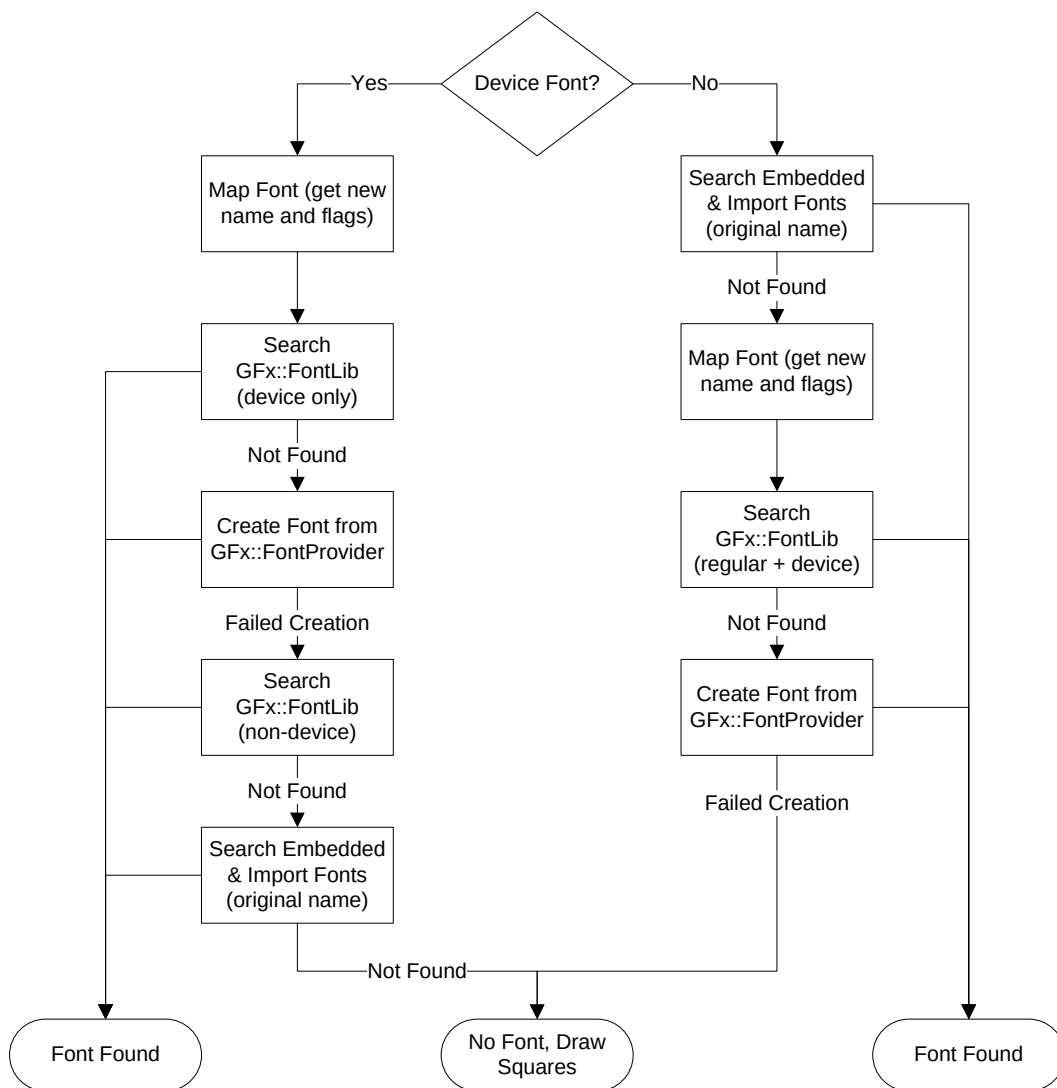
Scaleform

"

"

( , fall-back ). Scaleform

(null)





Gfx::FontLib    Gfx::FontProvider

## 4.2 Gfx::FontMap

Gfx::FontMap

3

```
[FontConfig "Korean"]
fontlib "fonts_kr.swf"
map "$TitleFont" = "Batang" Bold
map "$NormalFont" = "Batang" Normal
map "$SmallFont" = "Batang" Normal
```

"\$TitleFont"

C++

```
#include "Gfx/Gfx_FontLib.h"
. . .
Ptr<FontMap> pfontMap = *new FontMap;
Loader.SetFontMap(pfontMap);

pfontMap->MapFont("$TitleFont", "Batang", FontMap::MFF_Bold, scaleFactor = 1.0f);
pfontMap->MapFont("$NormalFont", "Batang", FontMap::MFF_Normal, scaleFactor = 1.0f);
pfontMap->MapFont("$SmallFont", "Batang", FontMap::MFF_Normal, scaleFactor = 1.0f);
```

"\$NormalFont"    "\$SmallFont"

MapFont

, MFF\_Original

```

        . Font
        scaleFactor

        Default 1.0f . Font

        Gfx::FontMap

        Gfx::MovieDef Gfx::Loader::CreateMovie

```

### 4.3 Gfx::FontLib

```

3 , Gfx::FontLib (1) "fonts_en.swf"
(2)

```

```

#include "Gfx/Gfx_FontLib.h"
...
Ptr<FontLib> fontLib = *new FontLib;
Loader.SetFontLib(fontLib);
fontLib->SetSubstitute("<default_font_lib_swf_or_gfx_file>");

Ptr<MovieDef> m1 = *Loader.CreateMovie("<swf_or_gfx_file1>");
Ptr<MovieDef> m2 = *Loader.CreateMovie("<swf_or_gfx_file2>");
...
fontLib->AddFontsFrom(m1, true);
fontLib->AddFontsFrom(m2, true);

```

```

AddFontsFrom . AddFontsFrom

AddRef

(

m1, m2).
/

```

, Gfx::FontLib

## 4.4 Gfx::FontProviderWin32

Gfx::FontProviderWin32 Win32 API ,  
GetGlyphOutline()

```
#include "Gfx/Gfx_FontProviderWin32.h"  
...  
Ptr<FontProviderWin32> fontProvider = *new FontProviderWin32(::GetDC(0));  
Loader.SetFontProvider(fontProvider);
```

Gfx::FontProviderWin32 Windows Display Context  
DC (::GetDC(0)).

## 네이티브 힌트 텍스트 사용

. Scaleform ,  
(CJK) ,  
CJK CJK  
API (Gfx::FontProviderWin32 Gfx::FontProviderFT2)

```
Font::NativeHintingRange vectorRange;  
Font::NativeHintingRange rasterRange;  
unsigned maxVectorHintedSize;  
unsigned maxRasterHintedSize;
```

Parameters vectorRange rasterRange

Font::DontHint -

Font::HintCJK - ,

Font::HintAll -

rasterRange vectorRange . maxVectorHintedSize

maxRasterHintedSize

SimSun

abcdefghijklmnopqrstuvwxyz vectorRange=GFxFont::DontHint  
ABCDEFGHIJKLMNOPQRSTUVWXYZ rasterRange=GFxFont::DontHint

倨偽僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞  
僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞

abcdefghijklmnopqrstuvwxyz vectorRange=GFxFont::HintAll  
ABCDEFGHIJKLMNOPQRSTUVWXYZ rasterRange=GFxFont::DontHint

倨偽僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞  
僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞

abcdefghijklmnopqrstuvwxyz vectorRange=GFxFont::HintAll  
ABCDEFGHIJKLMNOPQRSTUVWXYZ rasterRange=GFxFont::HintCJK

倨偽僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞  
僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞

"Arial Unicode MS"

abcdefghijklmnopqrstuvwxyz  
ABCDEFGHIJKLMNOPQRSTUVWXYZ  
倨偽僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞  
僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞

, Scaleform

Win32 FreeType

```
vectorRange = Font::DontHint;  
rasterRange = Font::HintCJK;
```

```
maxVectorHintedSize = 24;
maxRasterHintedSize = 24;
```

## 네이티브 힌트 설정

```
Gfx::FontProviderWin32 CJK
()
.
```

```
fontProvider->SetHintingAllFonts(. . .);
```

```
fontProvider->SetHinting(fontName, . . .);
```

```
SetHinting()          SetHintingAllFonts()
```

```
void SetHintingAllFonts(Font::NativeHintingRange vectorRange,
                        Font::NativeHintingRange rasterRange,
                        unsigned maxVectorHintedSize=24,
                        unsigned maxRasterHintedSize=24);
```

```
void SetHinting(const char* fontName,
                Font::NativeHintingRange vectorRange,
                Font::NativeHintingRange rasterRange,
                unsigned maxVectorHintedSize=24,
                unsigned maxRasterHintedSize=24);
```

```
fontName          UTF-8
.
```

## 4.5 Gfx::FontProviderFT2

```
David Turner          FreeType-2          . Gfx::FontProviderWin32
.
```

## FreeType

### MSVC

```
freetype<ver>.lib - DLL
freetype<ver>_D.lib - DLL
freetype<ver>MT.lib - ( CRT)
freetype<ver>MT_D.lib - ( CRT)
```

Where "<ver>" FreeType , 2.1.9 219 .

```
freetype2\include freetype2\objs include
```

Gfx::FontProviderFT2

```
#include "Gfx/Gfx_FontProviderFT2.h"
...
Ptr<FontProviderFT2> fontProvider = *new FontProviderFT2;
<Map Font to Files or Memory>
Loader.SetFontProvider(fontProvider);
```

FontProviderFT2(FT\_Library lib=0);

```
FreeType . 0 ( ), FreeType
```

Scaleform

FreeType

```
(FT_Done_FreeType ).
```

Gfx::Loader **길다**

```
FreeType (
```

```
). fontProvider-
```

```
>GetFT_Library() FT-Library FreeType
```

```

        , Gfx::FontProviderFT2
FreeType

```

## 폰트를 파일로 매핑

```

Win32 API      FreeType      (      )
        .      . Gfx::FontProviderFT2
        .
        .

void MapFontToFile(const char* fontName, unsigned fontFlags,
                  const char* fileName, unsigned faceIndex=0,
                  Font::NativeHintingRange vectorHintingRange = Font::DontHint,
                  Font::NativeHintingRange rasterHintingRange = Font::HintCJK,
                  unsigned maxVectorHintedSize=24,
                  unsigned maxRasterHintedSize=24);

"typeface"      "Times New Roman"      . "fileName"
"C:\\WINDOWS\\Fonts\\times.ttf"      . "fontFlags"
"Font::FF_Bold", "Font::FF_Italic"      "Font::FF_BoldItalic"      . "Font::FF_BoldItalic"
"Font::FF_Bold | Font::FF_Italic"      . "faceIndex"      FT_New_Face
0      0
        .
        .
        . Win32      , FreeType
        .

```

## 폰트를 메모리로 매핑

```

FreeType
        .
        .

void MapFontToMemory(const char* fontName, unsigned fontFlags,
                   const char* fontData, unsigned dataSize,
                   unsigned faceIndex=0,
                   Font::NativeHintingRange vectorHintingRange = Font::DontHint,
                   Font::NativeHintingRange rasterHintingRange = Font::HintCJK,
                   unsigned maxVectorHintedSize=24,
                   unsigned maxRasterHintedSize=24);

        (      )
        .

```

```
FILE* fd = fopen("C:\\WINDOWS\\Fonts\\times.ttf", "rb");
if (fd)
{
    fseek(fd, 0, SEEK_END);
    unsigned size = ftell(fd);
    fseek(fd, 0, SEEK_SET);
    char* font = (char*)malloc(size);
    fread(font, size, 1, fd);
    fclose(fd);
    fontProvider->MapFontToMemory("Times New Roman", 0, font, size);
}
```

( ).

naked

GFx::Loader

길어야

한다

Scaleform

FreeType

FreeType

```
Ptr<FontProviderFT2> fontProvider = *new FontProviderFT2;
fontProvider->MapFontToFile("Times New Roman", 0,
    "C:\\WINDOWS\\Fonts\\times.ttf");
fontProvider->MapFontToFile("Times New Roman", Font::FF_Bold,
    "C:\\WINDOWS\\Fonts\\timesbd.ttf");
fontProvider->MapFontToFile("Times New Roman", Font::FF_Italic,
    "C:\\WINDOWS\\Fonts\\timesi.ttf");
fontProvider->MapFontToFile("Times New Roman", Font::FF_BoldItalic,
    "C:\\WINDOWS\\Fonts\\timesbi.ttf");

fontProvider->MapFontToFile("Arial", 0,
    "C:\\WINDOWS\\Fonts\\arial.ttf");
fontProvider->MapFontToFile("Arial", Font::FF_Bold,
    "C:\\WINDOWS\\Fonts\\arialbd.ttf");
fontProvider->MapFontToFile("Arial", Font::FF_Italic,
    "C:\\WINDOWS\\Fonts\\ariali.ttf");
fontProvider->MapFontToFile("Arial", Font::FF_BoldItalic,
    "C:\\WINDOWS\\Fonts\\arialbi.ttf");

fontProvider->MapFontToFile("Verdana", 0,
    "C:\\WINDOWS\\Fonts\\verdana.ttf");
fontProvider->MapFontToFile("Verdana", Font::FF_Bold,
    "C:\\WINDOWS\\Fonts\\verdanab.ttf");
fontProvider->MapFontToFile("Verdana", Font::FF_Italic,
    "C:\\WINDOWS\\Fonts\\verdanai.ttf");
fontProvider->MapFontToFile("Verdana", Font::FF_BoldItalic,
    "C:\\WINDOWS\\Fonts\\verdanaz.ttf");
. . .
```



```
Loader.SetFontProvider(fontProvider);
```

```
. MapFontToFile()
```

## 제 부 폰트 렌더링 설정

Scaleform

( 2 ).

1.

/

2.

mip-mapped

gfxexport

3. Gfx::FontPackParams

4.

PC, Xbox 360 PS3

Wii

Gfx::FontProviderWin32

Gfx::FontProviderFT2

gfxexport

PSP

PS3

CPU

### 5.1 *(Glyph cache)*

Scaleform . Render::GlyphCacheConfig GFx::FontPackParams  
Renderer2D  
null , Scaleform

Scaleform

## 업그레이드 정보

Scaleform 3.x 4.0  
 . GfX 3.3 GfXLoader GfXFontCacheManager  
 , GfX 4.0 Renderer2D GlyphCacheConfig  
 . Scaleform GfX 4.0  
 5.2 .

5.2

```
Gfx::FontPackParams                                     'gfxexport'
```

'gfxexport'

```

bool Done = false;

if (Font has Static Textures with Packed Glyphs)
{
    if (GlyphSize <
        FontPackParams.TextureConfig.NominalSize * MaxRasterScale)
    {
        Draw the Glyph as a Texture using Static Cache;
        Done = true;
    }
}
else if (Dynamic Cache is Enabled AND
        GlyphSize < GlyphCacheParams.MaxSlotHeight)
{
    Draw the Glyph as a Texture using Dynamic Cache;
    Done = True;
}

if (Not Done)
{
    Draw the Glyph as Vector Shape;
}

```

## 5.2

```

renderer->GetGlyphCacheConfig()->SetParams(Render::GlyphCacheParams(0));

```

renderer

Render::Renderer2D

0

HAL

LRU ( )

"on the fly"

```
Render::GlyphCacheParams gcparams;  
gcparams.TextureWidth    = 1024;  
gcparams.TextureHeight   = 1024;  
gcparams.MaxNumTextures  = 1;  
gcparams.MaxSlotHeight   = 48;  
gcparams.SlotPadding      = 2;  
gcparams.TexUpdWidth     = 256;  
gcparams.TexUpdHeight    = 512;
```

```
renderer->GetGlyphCacheConfig()->SetParams(gcparams);
```

TextureWidth, TextureHeight -

2

MaxNumTextures -

MaxSlotHeight -

SlotPadding -

. Value 2

TexUpdWidth, TexUpdWidth -

256x512

(128K )

. 256x256

128x128

( )

UI

500

2,000

70-75%

| GFX, Dynamic cache                                                   | Readability                                                                          | Animation                                                                            |
|----------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <i>Anti-alias for readability</i><br><i>Anti-alias for animation</i> |  |  |

경고 폰트

힌트 문자가 없음

및

자동 힌트 사용 불가

"Zz"

"Basic Latin"

### 5.3 – *gfxexport*

*gfxexport* SWF GFX  
SWF  
DDS TGA -fc  
, *gfxexport*

| 명령줄 옵션      | 동작                                                                           |
|-------------|------------------------------------------------------------------------------|
| -fc         |                                                                              |
| -fcl <size> | 256<br>25% 1024 (Flash<br>)                                                  |
| -fcm        | FontCompactor<br>Boolean<br>10 ~ 70%<br>12(32 ) 16(64 ) +<br>12(32 ) 16(64 ) |

### 5.4 - *gfxexport*

-fonts , *gfxexport*  
Scaleform Player GFX

일반적인 응용 프로그램에 대해 내보내기 한 텍스처의 사용은 권장하지 않습니다

gfxexport

- 
- CPU
- Gfx::ImageCreator override  
Render::Renderer

mip-maps

'test.swf' 'test.gfx'

gfxexport -fonts -strip\_font\_shapes test.swf

-strip\_font\_shapes Scaleform

gfxexport

gfxexport

| 명령줄 옵션 | 속성 |
|--------|----|
| -fonts |    |



| 명령줄 옵션             | 속성                                                                      |
|--------------------|-------------------------------------------------------------------------|
|                    | ( )                                                                     |
| -fns <size>        | 48<br>Tri-linear mip-map                                                |
| -fpp <n>           | 3                                                                       |
| -fts <WxH>         | 256x256<br>( '-fts 128'<br>128x128. '-fts 512x128' )                    |
| -fs                |                                                                         |
| -strip_font_shapes | GFX                                                                     |
| -fi <format>       | <format> TGA8 ( ), TGA24 ( ), TGA32<br>DDS8<br>(-i ) TGA TGA8<br>DDS A8 |

경고

5.2

## 5.5

Font Glyph Packer

GfXExport

Basic Latin

Font Glyph Packer

. , Scaleform

```
Ptr<FontPackParams> packParams = *new FontPackParams();
Loader.SetFontPackParams(packParams);
```

```
GFXExport (          Scaleform      )
    "                "                .      GFXExport
```

```
(                )
```

```
Loader.GetFontPackParams()->SetUseSeparateTextures(Bool flag);
Loader.GetFontPackParams()->SetGlyphCountLimit(int lim);
Loader.GetFontPackParams()->SetTextureConfig(fontPackConfig);
```

```
SetUseSeparateTextures()
```

draw primitive

```
SetGlyphCountLimit()                .      0      "
```

```
"                .
                .      500
                (                )
```

```
SetTextureConfig()
```

```
FontPackParams::TextureConfig fontPackConfig;
fontPackConfig.NominalSize  = 48;
fontPackConfig.PadPixels    = 3;
fontPackConfig.TextureWidth = 1024;
fontPackConfig.TextureHeight = 1024;
```

NominalSize -

RAM

"NominalSize"

. NominalSize

" "

PadPixels -

1

TextureWidth, TextureHeight -

2

, Font Glyph Packer

GFX

- 1) `Ptr<FontPackParams> packParams = *new FontPackParams();`  
`Loader.SetFontPackParams(packParams);`  
`Loader.GetFontCacheManager()->EnableDynamicCache(false);`
- 2) `Ptr<FontPackParams> packParams = *new FontPackParams();`  
`Loader.SetFontPackParams(packParams);`  
`...`  
`renderer->GetGlyphCacheConfig()->SetParams(Render::GlyphCacheParams(0));`  
Font Glyph Packer 500

( )

## 5.6

Scaleform

2

Scaleform

Scaleform Player

CTRL + W

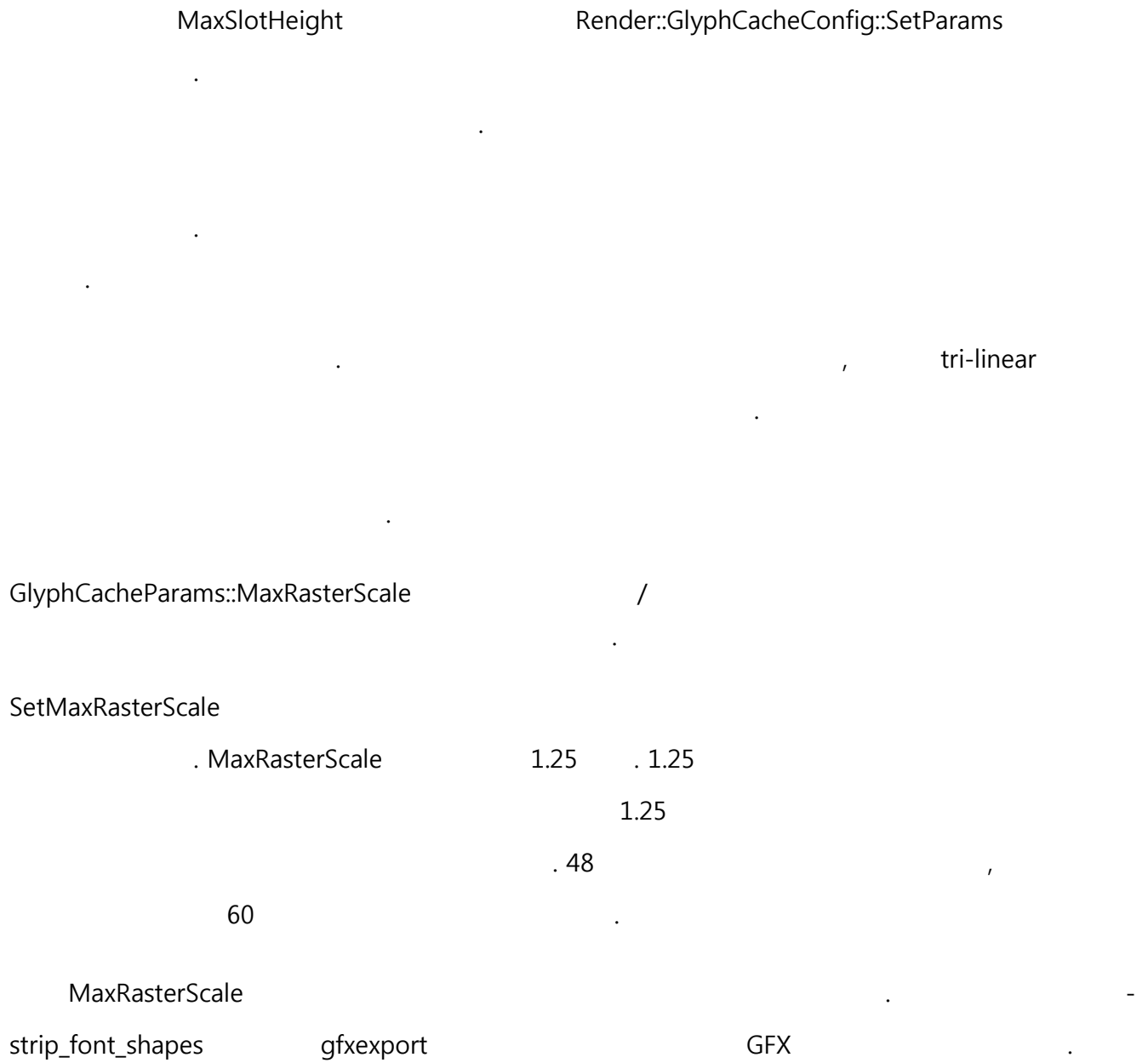
Small text is rendered with textures to allow for efficient rendering with one draw call. Large text, however, can be rendered as vectors allowing it to retain high quality.

Big Q  
@ % ?

Small text is rendered with textures to allow for efficient rendering with one draw call. Large text, however, can be rendered as vectors allowing it to retain high quality.

Big Q  
@ % ?

draw primitive



## 제 부 텍스트 필터 효과 및

## 확장

, Scaleform 2.x Blur, Drop Shadow Glow

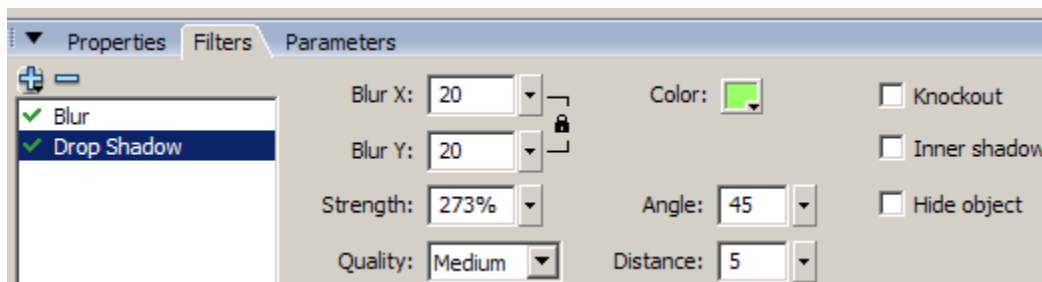
Scaleform

Blur, Drop Shadow Glow

Scaleform

Scaleform

### 6.1



Scaleform

Scaleform

Scaleform Blur Drop Shadow

. Glow

Drop Shadow

. Scaleform

Drop Shadow

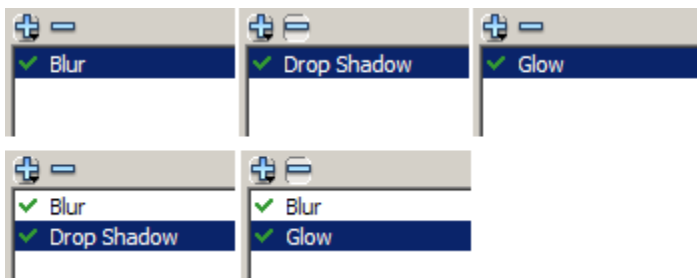
Glow

Blur

"Glow" "Drop Shadow"  
"Glow" "Drop Shadow"  
"Blur"  
"Blur" override

override

override



Drop Shadow Glow

Scaleform "shadow" "glow"  
"Blur" shadow  
"shadow" "glow" "Knockout" "Hide Object" (  
) "Inner Shadow" "Inner Glow" Scaleform

Scaleform

Blur X Blur Y

15.9 . Shadow Glow 1590%

"Knockout"

. Scaleform

" "

" "

Flash

GFx

*Text*

*Text*

"Knockout" "Glow" ( 0  
 "Drop Shadow" ) .

## 6.2

" "

3x3

3x3

Scaleform

. Scaleform

"Quality:Low"

Scaleform

( )

Scaleform

|       | Low quality | Medium quality | High quality |
|-------|-------------|----------------|--------------|
| Flash | <i>Text</i> | <i>Text</i>    | <i>Text</i>  |
| GFx   | <i>Text</i> | <i>Text</i>    | <i>Text</i>  |

Scaleform

" "

Scaleform

" " " "



<http://www.ph.tn.tudelft.nl/Courses/FIP/noframes/fip-Smoothin.html>

## 6.3

Scaleform ActionScript

Scaleform

( ) ,

/

( )

## 6.4 *ActionScript*

Scaleform AS2 AS3 / ( DropShadowFilter, BlurFilter, ColorMatrixFilter, BevelFilter) . Flash